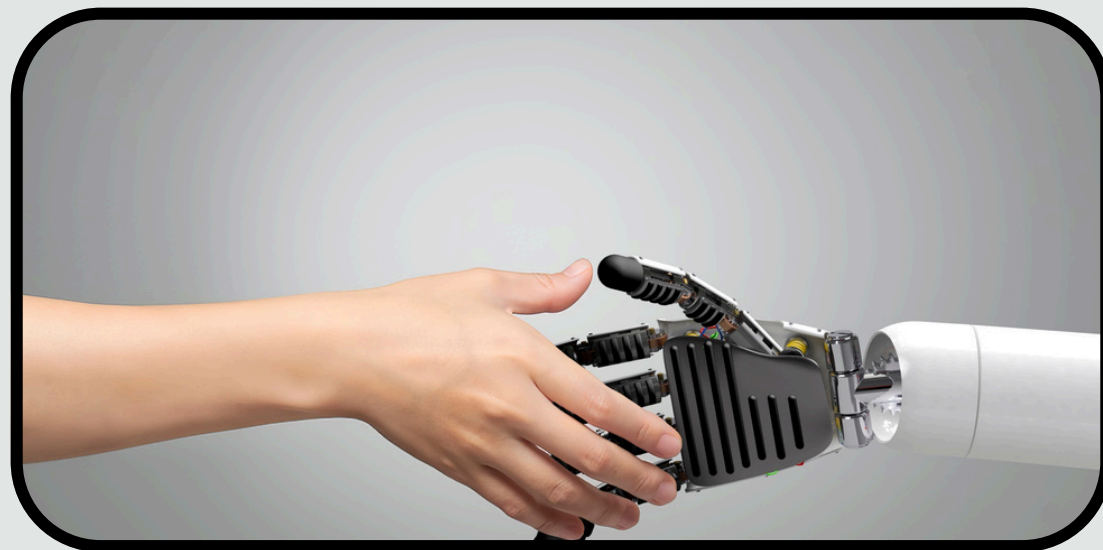


Gen AI module's Project



MDLM for Structured to Text Generation Tabular Data

A COMPARATIVE STUDY ON THE TOTTO DATASET



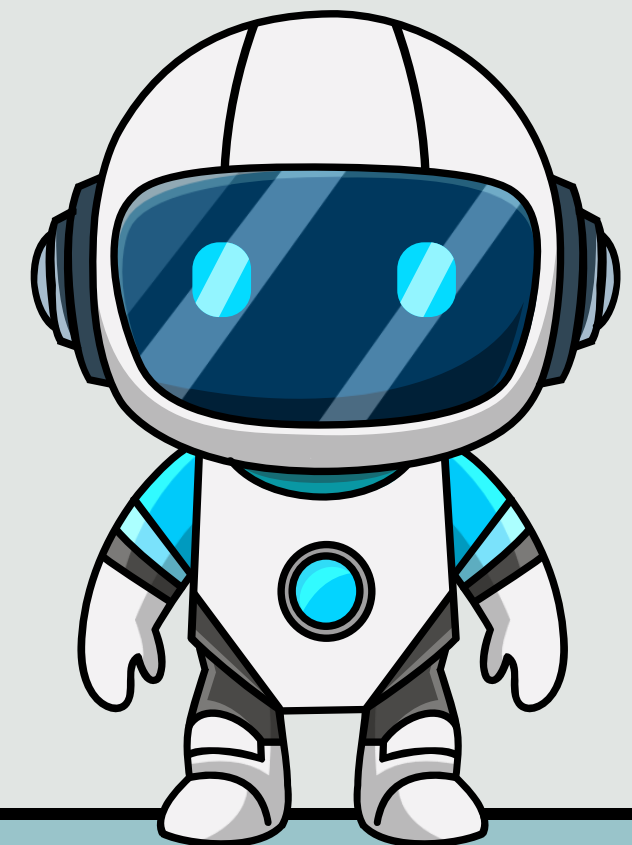
Presented by :

- Zaynab MERIMI
- Fatoumata WADIOU
- Noussayba EL MARRAKCHI
- Tasnim ATIG
- Sékou BAH
- Ahmed MAAOUIA

ING3 IA B

Professor :

- Arthur SATOUF



SUMMARY

- DATASET

-
- MDLM

-
- METRICS + COMPARAISON + MANUAL EVALUATION

-
- FINETUNING

ToTTo Dataset

[ToTTo Dataset Github](#)

Dataset Description

The ToTTo dataset consists of three `.jsonl` files, where each line is a JSON dictionary with the following format:

```
{
  "table_page_title": "'Weird Al' Yankovic",
  "table_webpage_url": "https://en.wikipedia.org/wiki/%22Weird_Al%22_Yankovic",
  "table_section_title": "Television",
  "table_section_text": "",
  "table": "[Described below]",
  "highlighted_cells": [[22, 2], [22, 3], [22, 0], [22, 1], [23, 3], [23, 1], [23, 0]],
  "example_id": 12345678912345678912,
  "sentence_annotations": [{"original_sentence": "In 2016, Al appeared in 2 episodes of BoJack Horseman .",
    "sentence_after_deletion": "In 2016, Al appeared in 2 episodes of BoJack Horseman as C",
    "sentence_after_ambiguity": "In 2016, Al appeared in 2 episodes of BoJack Horseman as I",
    "final_sentence": "In 2016, Al appeared in 2 episodes of BoJack Horseman as Captain Pe"}
}
```

Training subset: 5,000 examples

Evaluation subset: 500 examples

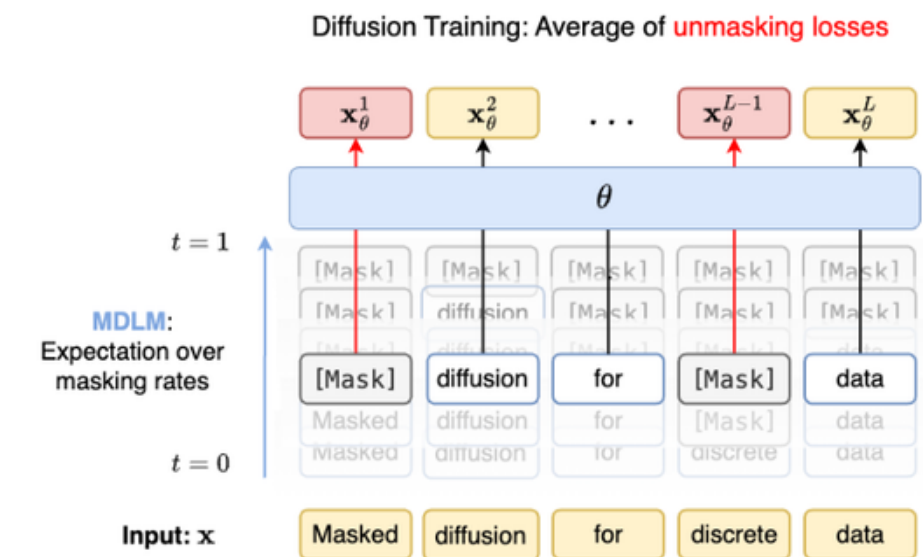
Masked Diffusion Language Model (MDLM)

Main points

- Non-autoregressive text generation
- Starts from a fully masked sequence
- Generates text through iterative denoising
- Uses bidirectional attention over the whole sentence
- Tokens are predicted in random order

Small comparison box

GPT-2	MDLM
Left-to-right generation	Iterative denoising
Sequential prediction	Bidirectional prediction
Fixed token order	Flexible token refinement



Why MDLM could Fit Table-to-Text

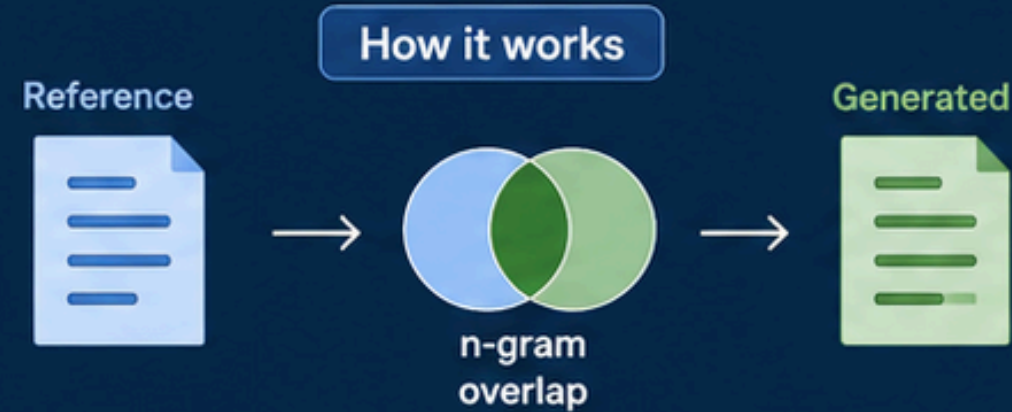
- Tables are inherently non-linear structures
- Information can be interpreted:
 - row-wise
 - column-wise
 - through highlighted cells
- MDLM captures relationships globally
- Better factual grounding in zero-shot setting
- Less dependent on sequential ordering

1. BLEU & ROUGE METRICS

Two standard metrics to evaluate text generation by comparing generated text with reference descriptions.

BLEU – Precision-Based Metric

Measures how many n-grams in the generated text also appear in the reference.



Example (unigrams & bigrams)

Reference: "The airport handled 10 million passengers."

Generated: "The airport served 10 million passengers."

Overlapping n-grams:

The airport 10 million million passengers

Interpretation

- ✓ High BLEU → generated text is close to the reference (good lexical precision)
- ✗ Low BLEU → wording differs significantly

Limitation



BLEU focuses on exact wording similarity and may penalize valid paraphrases.

ROUGE – Recall-Based Metric

Measures how much information from the reference is recovered in the generated text.



Variants

- ROUGE-1 → unigram overlap
- ROUGE-2 → bigram overlap
- ROUGE-L → longest common subsequence

Interpretation

- ✓ High ROUGE → important information from the table is preserved
- ✗ Low ROUGE → generated description misses relevant content

Why use both?

BLEU (Precision)

Checks how close the wording is to the reference.



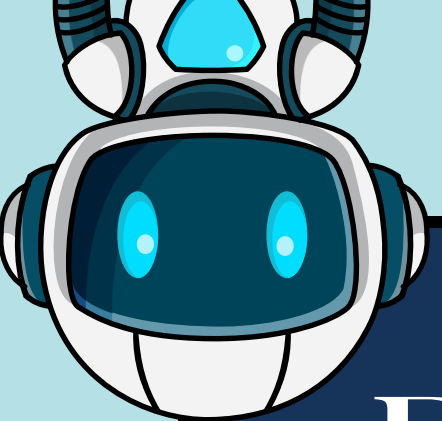
ROUGE (Recall)

Checks how much information is recovered.



Together, they provide a more complete evaluation of:

- textual fluency
- factual consistency
- information coverage



Performance comparison

Zero-shot (ZS) → MDLM dominates

- Higher scores in BERTScore
- Superior in BLUE and RED-L
- More consistent generations without examples

Few-shot (FS) → GPT-2

- Sometimes surpasses MDLM in BLUE and RED-L
- Effectively uses examples in context

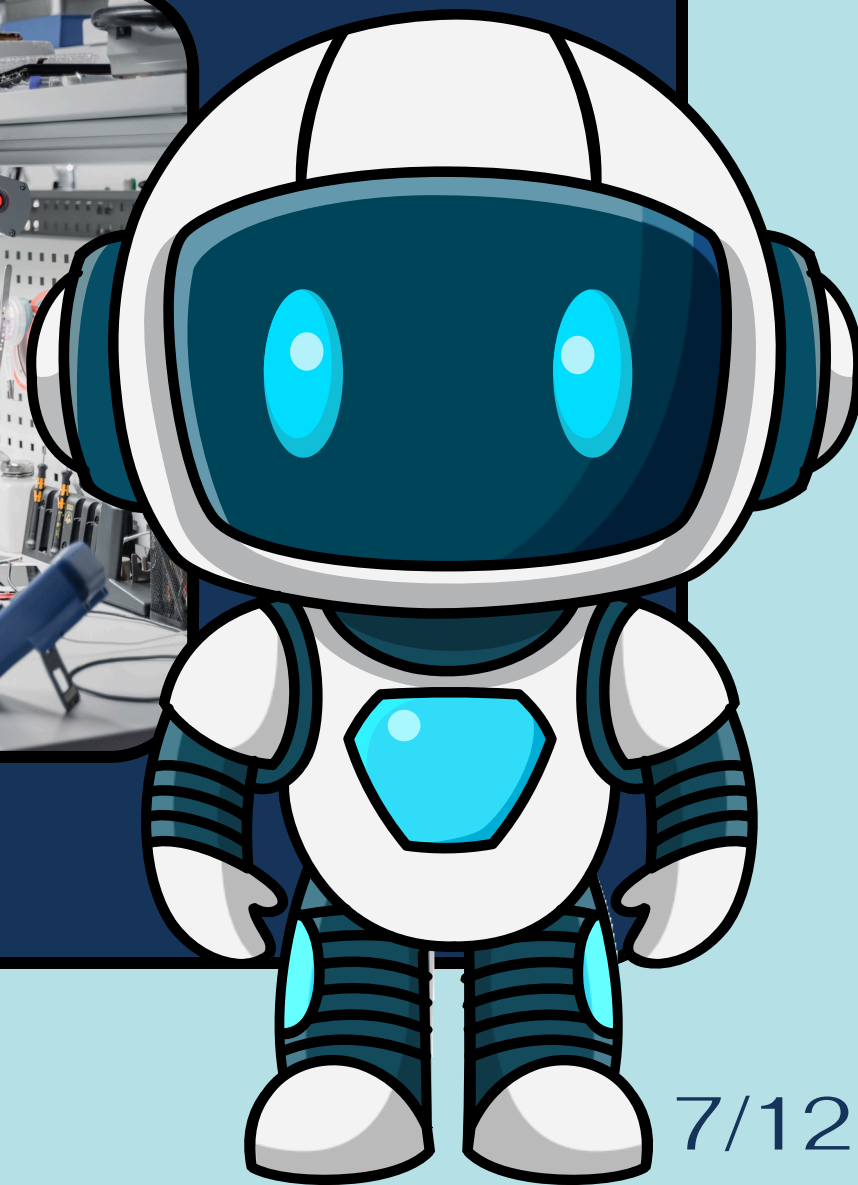
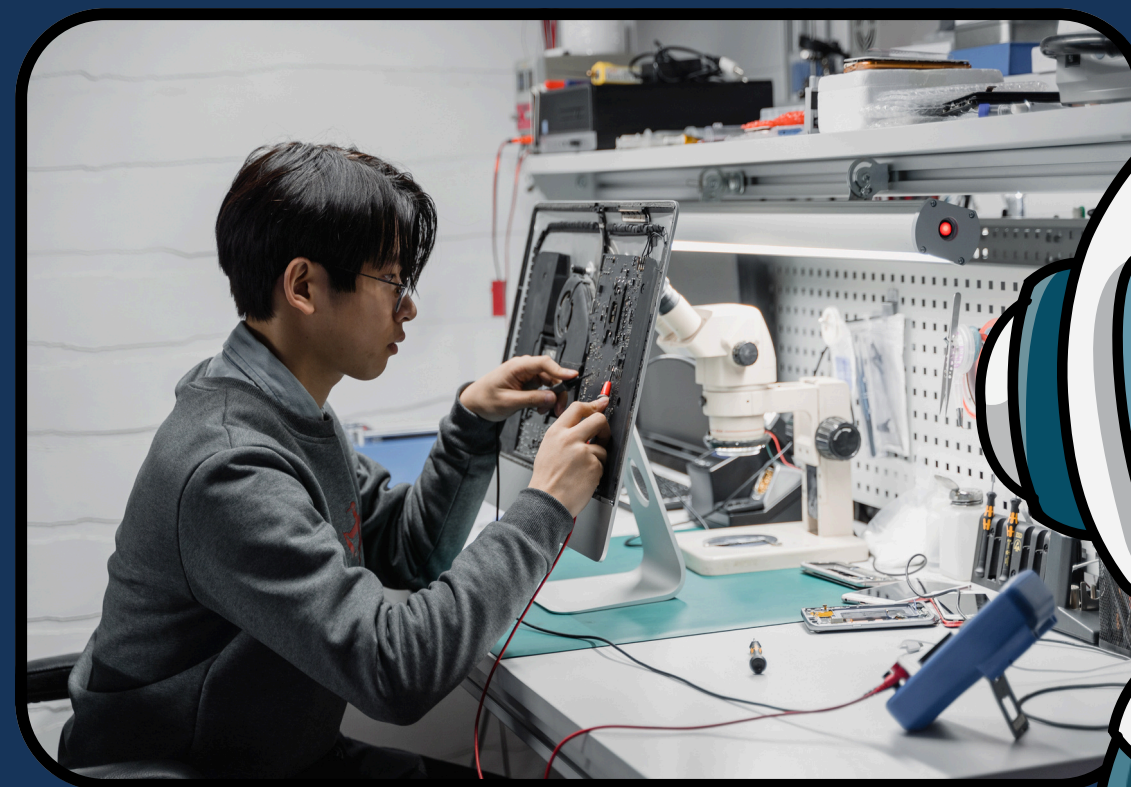
Diversity → Similar results

Métrique	GPT-2 (ZS)	MDLM (ZS)	GPT-2 (FS)	MDLM (FS)
BERTScore (+)	0.3886	0.5636	0.5231	0.5969
BLEU (+)	0.0001	0.0196	0.0221	0.0186
ROUGE-L (+)	0.0626	0.1955	0.2133	0.1696
Self-BLEU (-)	0.2109	0.2261	0.1394	0.1324

Manual Analysis

QUALITATIVE SCORING GRID

- **SCORE 2: FULLY FAITHFUL** - CORRECT ANCHOR TO TABLE
- **SCORE 1: PARTIALLY FAITHFUL** - DISTORTED FACTS
- **SCORE 0: HALLUCINATION** - OFF-TOPIC CONTEXT



Architecture-Specific Error Patterns

MDLM: FACTUAL GROUNDING

- **SUPERIOR ACCURACY:** ACTIVELY "READS" THE TABLE.
- **SEMANTIC ANCHOR:** ERRORS STAY TIED TO CORRECT ENTITIES, DATES, AND NAMES.
- **FORMAL FAILURES:** MECHANICAL REPETITIONS AND OVER-GENERATION ISSUES.
- **FIXABLE:** USABLE RESULTS VIA SIMPLE POST-PROCESSING (TRUNCATION).

GPT-2: DISCONNECTED FLUENCY

- **CRITICAL ISSUES:** SYNTACTICALLY FLUENT BUT FACTUALLY IRRELEVANT.
- **PURE HALLUCINATIONS:** ENTIRELY DISCONNECTED FROM INPUT DATA.
- **LEARNING GAP:** FAILS TO LEARN CONDITIONING BETWEEN STRUCTURE AND TEXT.
- **UNFIXABLE:** CANNOT BE CORRECTED BY SIMPLE POST-PROCESSING.

RESULTS MDLM

[illegible]

RESULTS GPT2

[illegible]

Zero-Shot vs. Few-Shot Dynamics

MDLM: THE "COGNITIVE NOISE" PENALTY

FEW-SHOT PROMPTING ACTUALLY **DEGRADES** PERFORMANCE.

- **ZERO-SHOT EXCELLENCE:** STRONG NATIVE UNDERSTANDING OF TABLES.
- **BIDIRECTIONAL RISK:** PROCESSES UNRELATED EXAMPLES AS DISTRACTIONS.
- **PRECISION DROP:** IRRELEVANT CONTEXT CAUSES A LOSS IN FACTUAL GROUNDING.

GPT-2: CONTEXT DEPENDENT

FEW-SHOT IS **VITAL** FOR AUTOREGRESSIVE BASELINES.

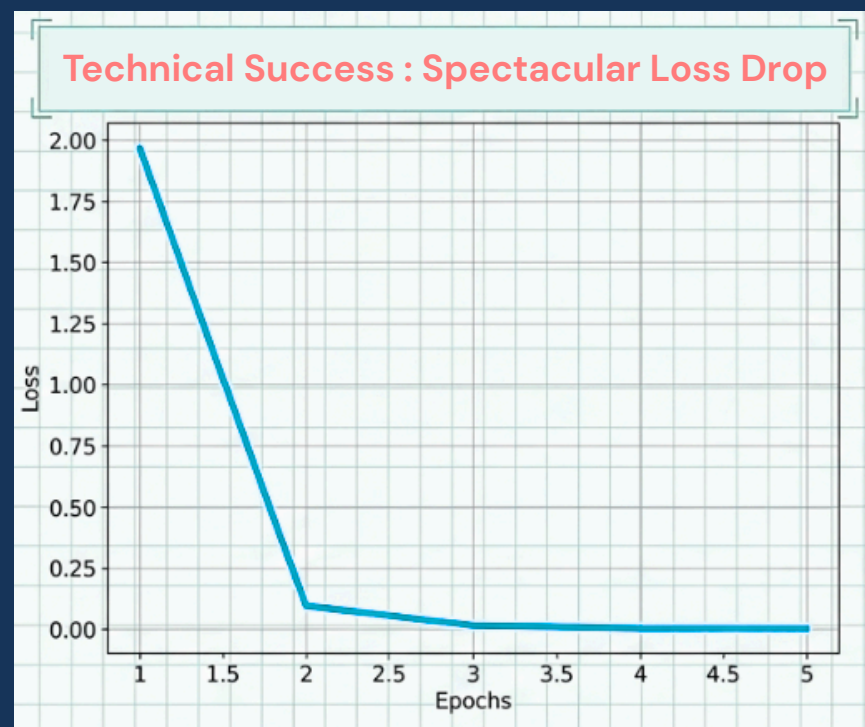
- **ZERO-SHOT:** COMPLETELY LOST; FAILS TO GRASP EXERCISE FORMAT.
- **FEW-SHOT:** MIMICS SYNTACTIC STRUCTURE EFFECTIVELY.
- **CAVEAT:** METRICS INFLATE (ROUGE/BLEU), BUT FACTUAL HALLUCINATIONS PERSIST.

FINETUNING : MDLM

Practical "Fail" : Metrics' values Drop (BLEU score)

Modèle Pré-Entraîné → ToTTo Dataset → Modèle Fine-Tuned

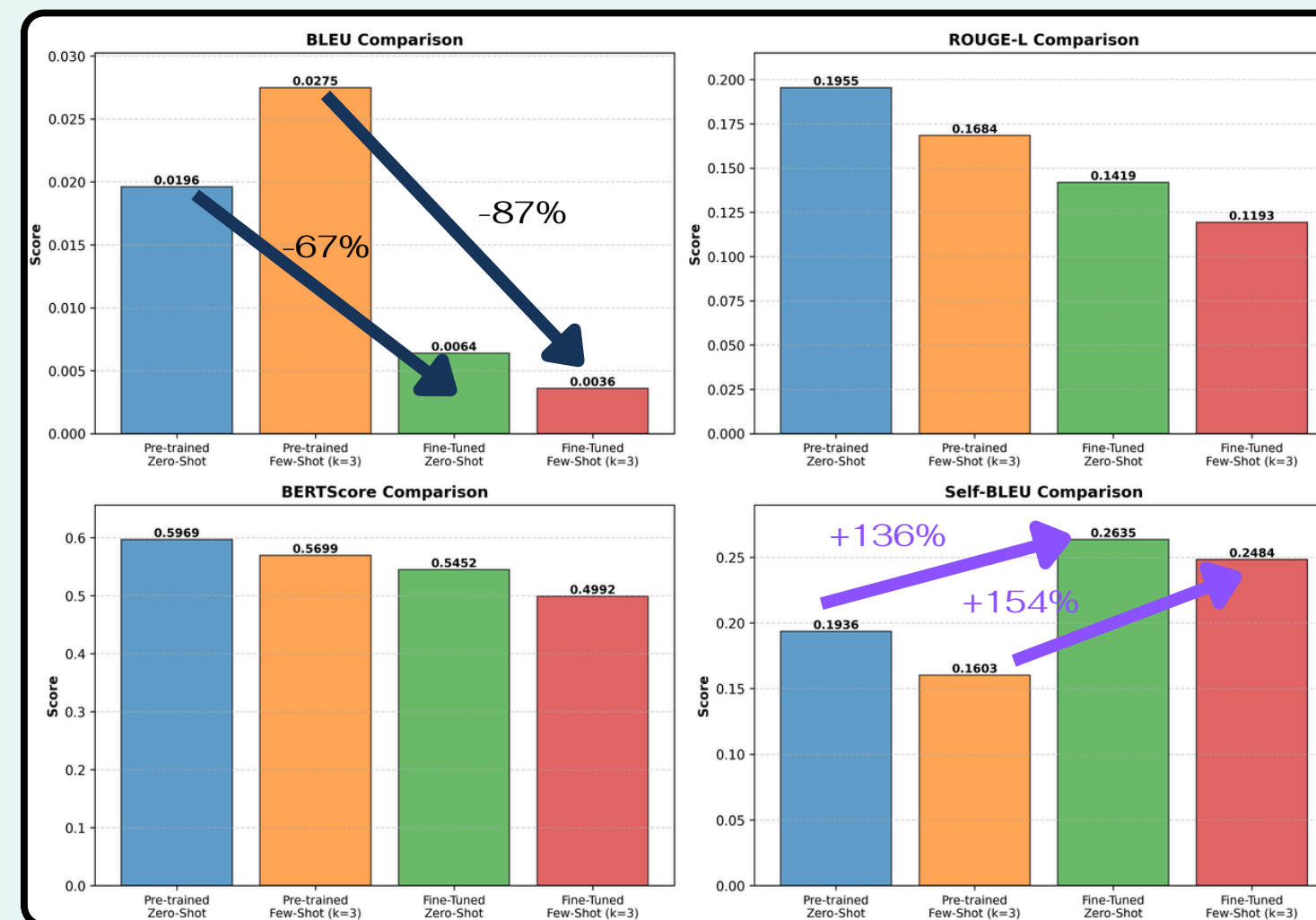
- Approach: Full fine-tuning (0.6B parameters) on a restricted subset (50 examples, 5 epochs)



VS

→ Full fine-tuning with the Totto dataset : reduced the performance and the MDLM's native generalization capabilities

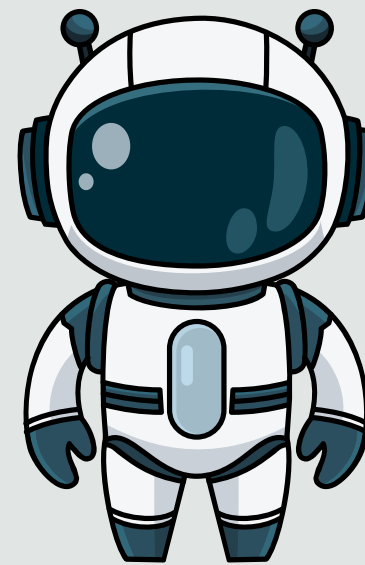
- The decrease in loss does not reflect improved understanding
- The model has overfit the dataset at the expense of factual accuracy :
 - BLEU score has **dropped** → **Worse lexical precision**
 - Rouge-L score has **roughly decreased** → **Almost same text recall performance**
 - BERTScore has remained **largely unchanged** → Maintained a **stable semantic similarity despite the drop in other metrics**
 - Self-BLEU has **boomed** → **Better similarity for multiple generations**



Conclusion & Future Work

GPT-2 (Autoregressive): Highly Context-Dependent

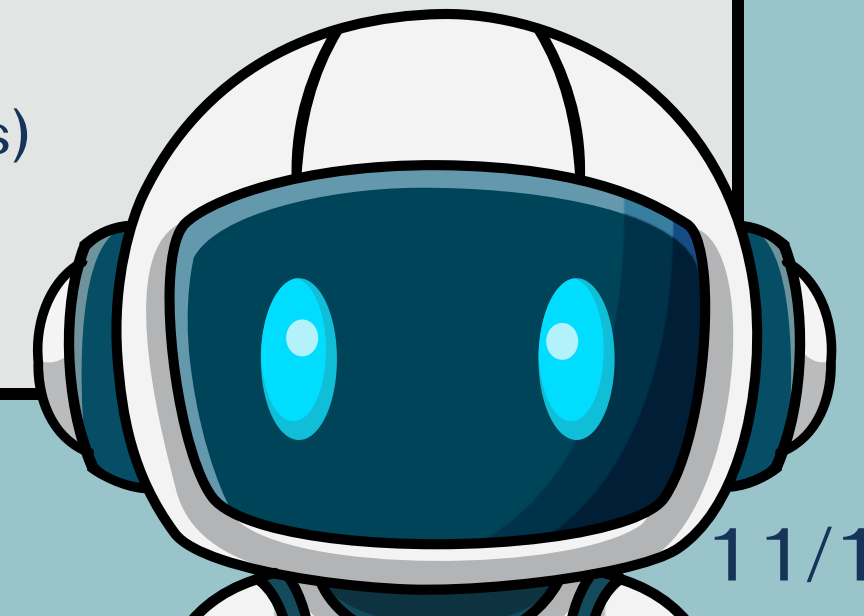
- **Zero-shot:** Struggles to parse non-linear tabular data.
- **Few-shot:** Drastic improvement (excellent at mimicking surface-level patterns).



MDLM (Diffusion): Strong Native Comprehension

- **Zero-shot:** Highest semantic similarity (BERTScore: 0.59). Outperforms GPT-2 natively.
- **Few-shot:** Slight performance drop due to attention dilution ("cognitive noise").
- **Fine-Tuning:** Full Fine-Tuning performance drops significantly due to OVERFITTING

- **Final Verdict:** Diffusion models are highly promising for non-linear tasks due to exceptional zero-shot generalization.
- **Future Work:** Execute the Full Fine-tuning with the train_subset (5000 examples) & Explore parameter-efficient fine-tuning (PEFT Like LoRA) and better context integration for diffusion models.



Thank you for
your attention!

